USING ENSEMBLE DISAGREEMENT TO STABILIZE CONFORMAL PREDICTION UNDER DISTRIBUTION SHIFT

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ABSTRACT

Using Ensemble Disagreement to Stabilize Conformal Prediction Under Distribution Shift

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Semantic segmentation of eelgrass (Zostera marina) from high-resolution drone-based imagery is essential for coastal habitat monitoring, restoration, and management, as these habitats see rapid changes due to climate change and human-induced influences. However, the reliability of generalizing these classification models not only relies on high-accuracy segmentation but also on rigorous uncertainty quantification that holds up when conditions change across years or locations. Conformal prediction (CP) converts a classifier's output into prediction sets with a guaranteed average coverage level for in-distribution data; however, the "vanilla" conformal score can under-cover in hard or out-of-distribution (OOD) regions under temporal drift. Motivated by difficulty-adaptive and heteroskedastic conformal approaches, this study investigates normalizing conformal scores by an epistemic proxy, ensemble disagreement. These scores re-scale nonconformity by a data-driven estimation of local difficulty learned on the calibration set, so that the global quantile is conservative where OOD risk is high. This study evaluates two normalization methods: a parametric linear normalization and a nonparametric normalization. This creates locally adaptive scores, which use ensemble variance as the proxy for local epistemic uncertainty. On drone imagery of the Morro Bay estuary, California (2018-2022), the models were trained on 2018-2021, calibrated on 2021, and evaluated on 2022 on OOD points. Normalized scores recover all or almost all of the lost coverage seen by vanilla split conformal prediction, increase the percentage of singletons, and reduce the spatial coverage variability. In this work, we provide a straightforward-to-implement drop-in framework for regaining coverage in challenging regions and out-of-distribution data, eliminating the need for retraining or labels at test time.

Keywords: conformal prediction, adaptive sets, uncertainty, eelgrass, remote sensing.

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Chapter 1

INTRODUCTION

Eelgrass (*Zostera marina*) is a seagrass species of temperate waters that provides an anchor to coastal ecosystems by providing sediment stabilization, carbon sequestration, habitats for protected species, and eliminating contamination. Warming oceans, storm regimes, and local disturbance have quickly reshaped the intertidal morphology and habitat extent. As these threats increase, there is a need to monitor changes to determine appropriate environmental management. In recent years, deep networks have made this intertidal mapping practical for complex imagery, achieving strong pixel-wise accuracy in areas such as Morro Bay, California (Tallam et al., 2023). However, the reliability of general utilization of these classification models not only relies on high-accuracy segmentation, but also on rigorous uncertainty quantification under temporal and spatial distribution shift. Imagery collected over different years, at different tides or seasons, can erode the model's calibration, causing it to be over-confident. This study builds off existing models to create a post-hoc uncertainty framework that is robust to distributional shift.

Uncertainty quantification is a requirement for flagging ambiguous areas. To address this, split conformal prediction provides an appealing framework. It offers a wrapping of any probabilistic predictor to produce prediction sets with an average coverage guarantee under exchangeability, simply by using a calibration set and a quantile nonconformity score (Angelopoulos & Bates, 2022). In classification, a natural choice of nonconformity score is:

$$s(x, y) = 1 - p_y(x) \tag{1.1}$$

where p_y is the model's predicted probability for the true class y. This yields a prediction set of all labels whose scores do not exceed a global threshold learned on calibration. With stable data, this can be simple and effective; however, in reality, the distribution of scores can differ by difficulty. A threshold calibrated over one year can underperform when a distributional shift occurs, such as temporal drift.

To remedy this, it is necessary to make the conformal prediction adaptive to drift or local difficulty. First, work in adaptive or locally weighted CP for regression has shown that normalizing scores by a measure of local difficulty can make a single global quantile more appropriate across heterogeneous inputs, which improves approximate conditional behavior (Dewolf, De Baets, & Waegeman, 2025; Gibbs & Candès, 2021; Tibshirani, Barber, Candès, & Ramdas, 2019). Second, research on uncertainty in deep learning has demonstrated that diversified deep ensembles provide practical, shift-aware signals of difficulty that tend to grow on out-of-distribution inputs (Lakshminarayanan, Pritzel, & Blundell, 2017). Using these ideas, this study implements a simple strategy for rescaling the classification score by a calibration-learned difficulty proxy, so that the global threshold becomes conservative under drift, without the need for recalibration or labels.

This thesis develops a rescaling design for eelgrass segmentation of multi-year drone imagery. The proposed normalizers are deliberately simple to adopt into existing pipelines. This includes:

1. a linear normalization $s' = (1 - p_y)/(1 + \lambda V)$ where V summarizes ensemble disagreement and λ controls degree of conservativeness
2. a learned normalizer $s' = s/\hat{a}(V)$ where $\hat{a}(V) = \mathrm{E}[s|V]$ is estimated on the calibration split via quantile binning and interpolation.

The impact of these designs is evaluated in Chapter 4 (Results), demonstrating improved robustness to drift while retaining the standard CP workflow. Importantly, neither approach requires labels at test time or retraining.

Chapter 2

BACKGROUND AND MOTIVATION

Effective eelgrass monitoring requires models that both provide accurate predictions, as well as how certain they are for those predictions. This section reviews the uncertainty concepts used in deep learning.

2.1 Uncertainty in Deep Learning

Uncertainty in predictive modeling is commonly broken up into two components: *epistemic* and *aleatoric*. These two components collectively comprise the total uncertainty in a model's predictions. Aleatoric uncertainty represents the noise inherent in observations, which is often impossible to remove, even with the addition of more data (e.g. sensor noise or true label ambiguity). Even a perfect model cannot remove it. Epistemic uncertainty represents the uncertainty of the model itself, which can be reduced with additional or higher-quality data (Kendall & Gal, 2017).

For modern vision models, raw probabilities are often miscalibrated, meaning the output confidence does not match empirical accuracy. This can be especially prevalent when the distribution of future data shifts, in which case models can be confident yet wrong. Post-hoc calibration, such as temperature scaling, rescales raw logistic output z via

$$\text{softmax}(z/T),$$

where $T > 0$ is a temperature parameter learned on a validation set. This improves in-distribution calibration without changing accuracy (Guo, Pleiss, Sun, & Weinberger, 2017). Under shift, however, calibration is not enough, as the structure of the errors can change.

Thus, it requires an uncertainty representation that integrates decision rules that signal the difficulty of the prediction.

2.2 Deep Ensembles

Deep ensembles can address uncertainty by aggregating predictions from models trained independently with different initializations or architectures. For estimating epistemic uncertainty, member disagreements can be calculated by considering the variance in member probabilities or logits. Given ensemble members $\{f_k\}_{k=1}^{K}$ that produce $p_y^{(k)}(x)$,

$$V(x) := Var_k\left(\left\{p_y^{(k)}(x)\right\}\right).$$

Ensembles have not only been found to be more robust under shift, but explicitly accounting for epistemic uncertainty helps prevent models from being overconfident when the input data changes (Ovadia et al., 2019). Additionally, ensembles require no specialized inference and can easily integrate with existing training code, providing a difficulty proxy for uncertainty. This does come at the additional computational cost of training additional models. However, many models are often trained during the model selection process, which, when ensembled, can outperform individual models.

2.3 Semantic Segmentation in Environmental Monitoring

Semantic image segmentation is a deep learning neural network technique that assigns class labels to each pixel within an image, delineating objects. Recent work demonstrates that deep networks can accurately classify intertidal eelgrass from drone imagery at useful resolutions (Tallam et al., 2023). Environmental monitoring can pose unique challenges, as not only are annotations costly and sometimes ambiguous, but the habitat is constantly shifting over seasonal and annual time scales. Even at the same site, small variations such as time of day, tidal height, etc., can alter the conditions of the imagery.

2.4 Conformal Prediction Foundations

Conformal prediction provides a model and distribution-free technique to convert probability scores into prediction sets with finite-sample marginal coverage under exchangeability (Angelopoulos & Bates, 2022). For a classifier with probabilities $\{p(x_i)\}_{i=1}^n$, a simple *nonconformity score* is $s(x) = 1 - p_y(x)$, which is small when the model is confident in the true class. In split conformal prediction, scores are computed on a calibration set C of size n and the finite-sample-corrected quantile is selected $\hat{q}_\alpha = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n}(\{s_i\})$, and the prediction set is defined as

$$C_\alpha(x) = \{y : s(x) \leq \hat{q}_\alpha\} \tag{2.1}$$

The full split conformal procedure is summarized in Algorithm 1.

Algorithm 1 Split Conformal Prediction for Classification

Input: Training data (X_{train}, Y_{train}), calibration data (X_{cal}, Y_{cal}), miscoverage level α

Output: Prediction set $C_\alpha(x)$ for each test point x

1. Train classifier f on (X_{train}, Y_{train}).
2. For each calibration point (x_i, y_i):
   (a) Compute predicted probabilities $\hat{p}(x_i) = f(x_i)$.
   (b) Compute nonconformity score
   $$s_i = 1 - \hat{p}_{y_i}(x_i).$$
3. Compute the finite-sample corrected quantile
   $$\hat{q}_\alpha = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n}(s_1, \dots, s_n).$$
4. For a new input x:
   (a) Compute probabilities $\hat{p}(x) = f(x)$.
   (b) Return prediction set
   $$C_\alpha(x) = \{y : 1 - \hat{p}_y(x) \leq \hat{q}_\alpha\}.$$

This procedure guarantees $P\{Y \in C_\alpha(X)\} \geq 1 - \alpha$ for new points drawn from the same distribution as the calibration data regardless of the underlying model. Two limitations of conformal prediction motivate this thesis. The first limitation is that the guarantee is marginal. There is no guarantee of conditional coverage for specific subsets of the data. This means that coverage may fall below the target for a specific class, for certain spatial regions of the data, or for more difficult inputs. Second, conformal prediction has mathematical validity due to the concept of symmetry (data could have been seen in any order) as long as the data is exchangeable. Coverage guarantees are lost as distribution drifts, and the model can become overly confident, leading to under-coverage. To address these problems, we need to use an adaptive variant of conformal prediction, which is more conservative for harder inputs.

2.5 Adaptive or Shift-Aware CP

There is a substantial amount of literature that aims to make CP adaptive to heterogeneity or robust during dataset drift. A recurring issue is that a single global conformal threshold calibrated on held-out in-distribution data can be both overly conservative for easy inputs and insufficiently conservative for difficult ones. This problem becomes even more pronounced for a model deployed under dataset drift.

In regression, heteroskedasticity motivates methods that adapt to local noise, instead of constant-width thresholds. For example, conformalized quantile regression (Romano, Patterson, & Candès, 2019), combines conformal prediction with classical quantile regression to provide intervals that vary with conditional spread. More generally, normalization approaches target a similar idea, rescaling nonconformity scores by an estimate of uncertainty so a global quantile is more appropriate under heterogeneity (Dewolf et al., 2025).

When exchangeability is violated between the test and calibration set, achieving reliable coverage can be even more challenging. Methods such as adaptive conformal inference (Gibbs & Candès, 2021) use an adaptive time-varying parameter that is continuously re-estimated to maintain coverage. This method requires online feedback. However, in our application labels are obtained through manual annotation, which occurs offline with significant delay.

Other approaches specifically target covariate shift, such as Tibshirani et al. (2019), which uses weighting based on the likelihood ratio between test and calibration distributions. However, this requires estimating the density ratio $p_{\text{test}}(x)/p_{\text{train}(x)}$, which can be challenging to estimate reliably in high-dimensional imagery.

While these and other methods can provide formal routes for validity beyond exchangeability, they often require continuous feedback, additional assumptions, or quantities that are difficult to estimate reliably in high-dimensional vision problems.

In this study, we focus on difficulty-normalized conformal scores using epistemic uncertainty. We do not focus on establishing conformal guarantees under shift, and instead tailor the idea of difficulty adaptive/normalized scores to temporal drift.

Let $S = s(X, Y)$ denote the standard noncomformity score and let $V = v(X)$ be a one-dimensional difficulty proxy from the deep ensemble (member-probability variance after temperature scaling). We can use this proxy to stabilize the score with respect to V. We normalize under a multiplicative difficulty assumption

$$S = a(V)U, \quad a(V) > 0, \quad U \perp\!\!\!\perp V \tag{2.2}$$

where U represents a difficulty invariant component of the nonconformity score, assumed to be independent of V. Then, $S' = S/a(V) \approx U$, so $S'|V$becomes approximately invariant, and locally tuned to difficulty. Normalization targets the actual failure under shift, so that the global threshold will become too lenient for the hard OOD inputs. On the calibration set, we estimate $\hat{a} \approx E[S|V = v]$ via quantile binning and smooth interpolation, and then conformalize the normalized score $S' = S/\hat{a}(V)$. Thus, if the conditional distribution

$S|V = v$ is roughly the same across years:

$$S \mid V = v(calibration) \approx S \mid V = v(test),$$

then we expect the normalized scores to be approximately stable conditional on V. Then the global quantile will be appropriate across the shifted years. Intuitively, pixels with large member disagreements (difficult pixels) would have larger scores. Dividing by $\hat{a}(V)$ will equalize the scores, so that difficult and easy pixels are all scored equally, being precise on easy inputs and conservative on difficult ones. By using a difficulty signal (epistemic uncertainty) that is specifically "shift-aware," we expect that the score becomes shift-aware as well.

Chapter 3

METHODOLOGY

3.1 Data

This study uses imagery of the estuary in Morro Bay, California. The data was collected between 2018 and 2022. Eelgrass was mapped using a DJI Phantom 4 Pro drone with a high-resolution camera. Data was collected between the months of November and January (of the following year) during days of extreme low tides, when eelgrass is exposed. Flight times were approximately 1 hour due to limitations of tide duration. High-resolution estuary-wide orthomosaics were acquired (~5000 images per year) and manually annotated to delineate between eelgrass and non-eelgrass polygons by a team of GIS specialists, as mentioned in Tallam et al. (2023). This includes iterative validation checks, resulting in ~1000 hours spent per year. The resulting polygons were rasterized to match imagery, resulting in pixel-wise labels for semantic segmentation.

For model development, imagery from 2018-2021 is treated as in-distribution (ID), while imagery from 2022 is treated as temporally out-of-distribution (OOD). For 2022, instead of fully labeling the orthomosaic (which, as mentioned, can be expensive), a set of 1000 randomly sampled points was human labeled as either eelgrass or non-eelgrass. The 2018-2021 data was tiled into chips of size 448x448 pixels and a stride of 224 pixels, resulting in 50% overlap between chips.

Figure 3.1: Example 448×448 px tile from the drone orthomosaic (left) with the corresponding manually digitized eelgrass segmentation mask (right).

2021 was chosen as the *calibration year* as it is the most recent pre-2022 year. For 2021, the data was partitioned into four subsets:

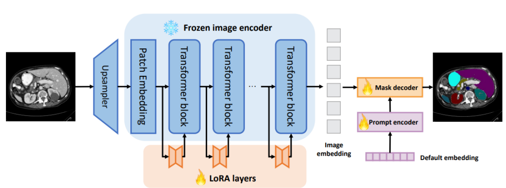
1. a train subset (together with 2018-2020)
2. a small temperature-scaling subset to estimate the models' over- or underconfidence in their softmax probabilities
3. a CP calibration subset to compute conformal scores and quantiles
4. an evaluation subset to test in-distribution coverage. The OOD 2022 evaluation set is thus never used in training, temperature scaling, or conformal calibration.

3.2 Model Training and Inference Pipeline

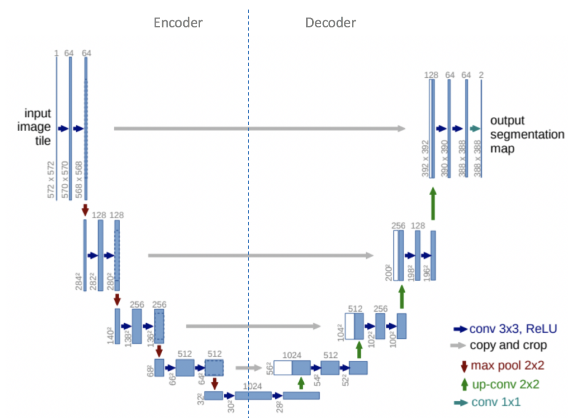
3.2.1 Segmentation Ensemble

To obtain the epistemic uncertainty proxy and achieve high-accuracy predictions, we trained a heterogeneous deep ensemble of semantic segmentation models to increase model diversity. In total $K = 6$ models were added to the ensemble. Of the 6, 2 of each model architecture were added. These architectures include DeepLabv3+, UNet, and two variants of Segment Anything Model (SAM) that were fine-tuned with low-rank adapters (Sam-LoRA). Each model type had one multi-year (2018-2021) model in the ensemble and a single-year (2021 only) model. Training was performed using dual NVIDIA RTX 3090 (24 GB VRAM) GPUs, and took approximately one week for 4-year models and three days for single-year models.

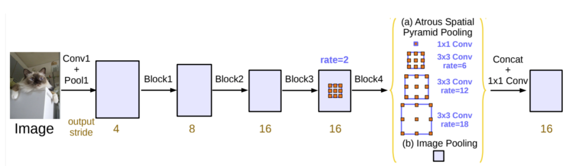
The models were trained using the ArcGIS Learn Python module. For the two DeepLabV3+ models, we used a ResNet-101 backbone pre-trained on ImageNet, whereas the U-Net models, by default, use a ResNet-34 encoder. SAM-LoRA uses a pre-trained SAM encoder (vision transformer-based model) and inserts low-rank adapter layers to adapt the model for eelgrass segmentation, without fully tuning the model.



(a) SAM-LoRA architecture



(b) U-Net architecture



(c) DeepLabv3+ architecture

Figure 3.2: Model architectures included in the ensemble. Images from the ArcGIS Learn developer guides.

Models were trained for 20 epochs, and using a held-out validation set, the best checkpoint (best validation loss) was selected to be included in the ensemble.

3.2.2 Probability Calibration and Aggregation

At inference time, each of the models produces per-pixel log-odds, which are then temperature-scaled, converted to class probabilities, and aggregated. Temperature scaling calibrates the model confidence using a scalar temperature $T > 0$, which is calculated by minimizing the negative log-likelihood on a holdout set designated for calibration (Guo et al., 2017). For each model f_k, producing log-odds $z^{(k)}(x)$, estimated probabilities are then computed using temperature scaled softmax:

$$\hat{p}^{(k)}(x;T) = \text{softmax}\left(\frac{z^{(k)}(x)}{T}\right).$$

We then compute the ensemble-averaged calibrated probability for class y at pixel x,

$$\hat{p}(y|x) = \frac{1}{K}\sum_{k=1}^{K} p_k(y|x;T).$$

The ensemble prediction is thus the aggregated probability. Each model's individual probability output, $p_k(y|x;T)$, is also retained for computing the epistemic uncertainty proxy.

For each labeled point x, we compute the epistemic variance proxy based on member disagreement for the true class,

$$\hat{V}(x) = \widehat{Var}\{p_k(y_{true}|x;T)\}$$

To assess how the uncertainty proxy relates to errors, we use selective accuracy curves and correlation analyses. This was done with the variance scores (our epistemic proxy), as well as entropy (1) and mutual information (2) uncertainty scores for comparison. Let

$$H(p) = -\sum_{y} p(y)\, \log p(y)$$

represent the entropy of a class-probability distribution p. Then the comparison metrics are

$$1.H(\widehat{p}(y \mid x))$$

and

$$2.\mathrm{MI}(x) = H(\widehat{p}(y \mid x)) - \frac{1}{K}\sum_{k=1}^{K} H(p_k(y \mid x))$$

However, these were not tested in conformalizing equations as entropy mixes aleatoric and epistemic uncertainties and is implicitly correlated with the probability-based nonconformity scores s_y, and mutual information seemed to rank errors less effectively according to selective accuracy analysis. To create a selective accuracy curve, take a given uncertainty score $u(x)$ and sort by decreasing $u(x)$. We then iteratively recompute accuracy scores on smaller and smaller subsets of higher confidence points. For an informative uncertainty metric (that ranks errors well), we should observe a monotonic increase in accuracy as the size of the subset decreases.

3.3 Split CP for Classification

The ensemble model is treated as a single probabilistic classifier for conformal prediction. We apply split conformal prediction to obtain per-pixel prediction sets with finite-sample coverage under exchangeability (Angelopoulos & Bates, 2022). For each labeled pixel x and class y, we define a nonconformity score such as the vanilla $S(x, y) = 1 - \widehat{p}(x)$, which is large when the model assigns a low probability to the true class (nonconforming). On the 2021 calibration subset C of size n, we compute nonconformity scores on all pixels. Then we sort the pixels and compute the finite-sample corrected empirical $(1-\alpha)$ quantile $\widehat{q}_\alpha = \text{Quantile}_{\lceil (n+1)(1-\alpha)\rceil/n}(\{s_i\})$ for a target miscoverage level. For a new pixel, the CP set is

$$C_\alpha(x) = \{\, y \in \{\text{eelgrass, background}\} : s(x, y) \leq \widehat{q}_\alpha \,\}\,.$$

Under the assumption that the calibration and test data are exchangeable, marginal coverage is guaranteed: $\Pr\{\, Y \in C_\alpha(X) \,\} \geq 1-\alpha$. CP sets are computed for both a held-out 2021 ID set and the 1,000 labeled 2022 OOD points. We used the vanilla score to provide a baseline for OOD data coverage compared to the variance-aware methods.

3.4 Variance-Aware Score Normalization

As discussed in the background, split CP can under cover in difficult or OOD regions since the score distribution will be larger where the classifier is less reliable. We address this by normalizing by the difficulty proxy, ensemble variance.

Let $V(x)$ denote the ensemble variance at pixel x. For each calibration point x_i, we observe a pair (s_i, v_i). We normalize the score according to Equation 3.1:

$$S^\star = \frac{S}{g(V)} \tag{3.1}$$

where $g(\cdot)$ is a positive scaling function learned on the calibration set. Split CP is then applied to $s_i^\star = s_i/g(v_i), i \in C$, which yields a new threshold $\widehat{q}_\alpha^\star$ and new prediction sets

$$C_\alpha^\star(x) = \{\, y : s^\star(x,y) \leq \widehat{q}_\alpha^\star \,\}\,.$$

Intuitively, where v(x) is large (ensemble model disagrees; model is uncertain), $g(v)$ is larger and thus $s^\star$ shrinks. This makes it more lenient in hard/OOD regions, reducing under-coverage. Two choices of g were tested, a simple parametric linear variance scaler and a nonparametric normalization based on the relationship between S and V. For both methods, we first rescale the ensemble variance $v(x)$ using the minimum and maximum values observed on the calibration set so that the resulting values lie approximately in $[0,1]$. Specifically, we define

$$\tilde{v}(x) = \frac{v(x) - v_{\text{min}}}{v_{\text{max}} - v_{\text{min}}},$$

where

$$v_{\text{min}} = \min_{i \in \mathcal{C}} v(x_i), \qquad v_{\text{max}} = \max_{i \in \mathcal{C}} v(x_i).$$

3.4.1 Parametric Linear Scaling

For the parametric version, we create a simple relationship between variance and shrinkage:

$$S_\lambda(x, y) = \frac{S(x, y)}{1 + \lambda V(x)} \tag{3.2}$$

λ controls how aggressively we shrink the score in high-variance regions. For $\lambda = 0$, S_λ simply reduces to the vanilla score. In general, λ is optimized via a grid search. In this application, we used a small manual grid search of $\lambda \in 0, 0.5, 1, 2, 4$. λ was chosen to optimize a balance of two metrics: 1. improving OOD coverage, and 2. retaining a high percentage of singletons (more on CP evaluation in Section 3.5). This method remains a simple "drop-in" method on top of vanilla split CP.

3.4.2 Nonparametric Normalization

The parametric shrink assumes a linear relationship between predictive difficulty and the appropriate scale correction. This does not necessarily match empirical behavior. To account for this, we also tested a nonparametric normalization that estimates the correction function $a(v)$ from the data. We assume a multiplicative difficulty model as defined in Equation 2.2, where $a(V)$ captures the difficulty-dependent scaling of the score. Under this assumption, dividing by $a(v)$ should remove most of the dependence of S on V. On the calibration set, we estimate $a(\cdot)$ using a smoothed estimate of the conditional mean $E[S|V = v]$. This is done by first partitioning the calibration set into $B = 20$ bins $B_1, \ldots, B_B$ based on quantiles of V, the ensemble variance of the per-model predicted probabilities $\{p_k(y_{true}|x; T)\}$. Thus, each bin has roughly an equal number of points. For each bin, we then compute the mean variance v_b and the mean score $\bar{s}_b$. These $(v_b, \bar{s}_b)$ pairs summarize the typical score magnitude at difficulty around v_b. We then construct a continuous estimate $\hat{a}(v)$ by interpolating linearly between the bin centers, treating v_b as x-values and $\bar{s}_b$ as y-values, yielding a function defined for all v.

The key assumption is that the relationship between score and difficulty is approximately invariant across the adjacent years. If $S_{2021}|V_{2021} = v \approx S_{2022}|V_{2022} = v$, then the single global quantile behaves as if locally tuned to difficulty in both the calibration year and the shifted year.

3.5 CP Evaluation

To evaluate the performance of vanilla and variance-aware CP under temporal drift, two main metrics were evaluated with a few complementary metrics. The first, and most important, evaluating metric is global marginal coverage. This can simply be calculated as the proportion of evaluation sets including the ground truth:

$$\text{Coverage} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{\, y_i \in C(x_i)\,\}$$

This is then compared to the target $1 - \alpha$ coverage. The next metric to examine is the set composition. For each pixel, the prediction set can be empty ($\emptyset$), a singleton ({eelgrass} or {other}), or the full set ({eelgrass, other}). We report the proportion of each category and can use this to measure the efficiency of the sets. A good conformal model in binary settings aims to maximize the number of singletons while retaining coverage $\geq$ target. This evaluation method can be extended to the multiclass setting, where predictions may contain more than two labels and efficiency is instead measured by *minimizing* the average set size. In the binary case, set composition provides the most interpretable results, as the empty and two-set cases both represent two extremes of uncertainty, while singletons correspond to the most informative predictions.

Beyond the main evaluation metrics, we added a few complementary metrics to evaluate each method. The first is class-conditional coverage. To assess whether coverage is balanced between eelgrass and other, we compute the coverage for each individual class:

$$\text{Coverage}_y = \frac{1}{N_y} \sum_{i=1}^{N_y} \mathbf{1}\{\, y_i \in C(x_i) \,\}, \quad y \in \{\text{eelgrass, other}\}$$

Next, to examine the effectiveness of normalization, we examine the coverage vs. difficulty. We bin observations into quantiles of V and compute coverage within each bin to produce coverage vs. variance curves. Lastly, we aimed to evaluate not only how the methods handle temporal drift but also spatial heterogeneity, such as areas of the estuary being more difficult to classify than others. Spatial robustness was assessed by partitioning the 2022 evaluation points into spatially localized blocks. Points were projected to planar coordinates (meters) and ordered using Morton (Z-order) space-filling curve to preserve spatial locality. We partitioned the ordering into $B = 10$ equal-sized blocks (n=100), and computed coverage within each block. We summarize spatial equity by the standard deviation of block-wise coverage and by the correlation between block-wise coverage and block-wise mean V. More metrics and visuals were also created that will be included within the results; however, these capture the main evaluation metrics.

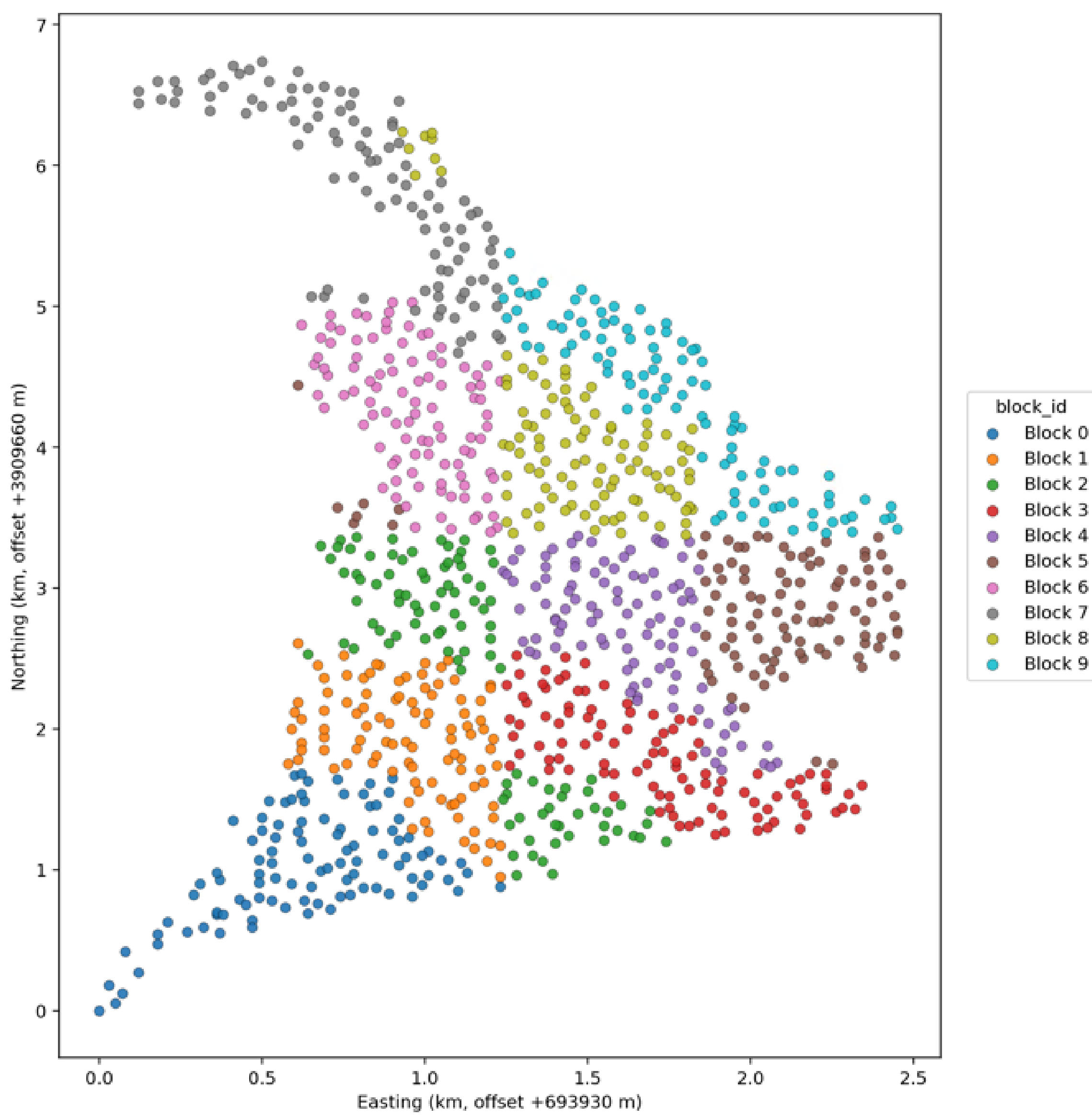


Figure 3.3: Spatial blocks (10 equal-count spatial blocks formed by Morton ordering)

Chapter 4

RESULTS

4.1 Ensemble Results

Figure 4.1 shows a representative 2022 chip with the ensemble outputs. This includes the raw chip imagery, the ensemble mean probability for the target class, the ensemble variance, and the final class segmentation. Visually, it seems that high ensemble variance aligns with boundaries between eelgrass and non-eelgrass patches as well as ambiguous pixels, such as darker shadows or where features resemble eelgrass RGB. In the example chip, it can be clearly seen that the variance spikes at the boundary between eelgrass and non-eelgrass patches. These results suggest that the disagreement signal is capturing the genuinely difficult pixels in the imagery.

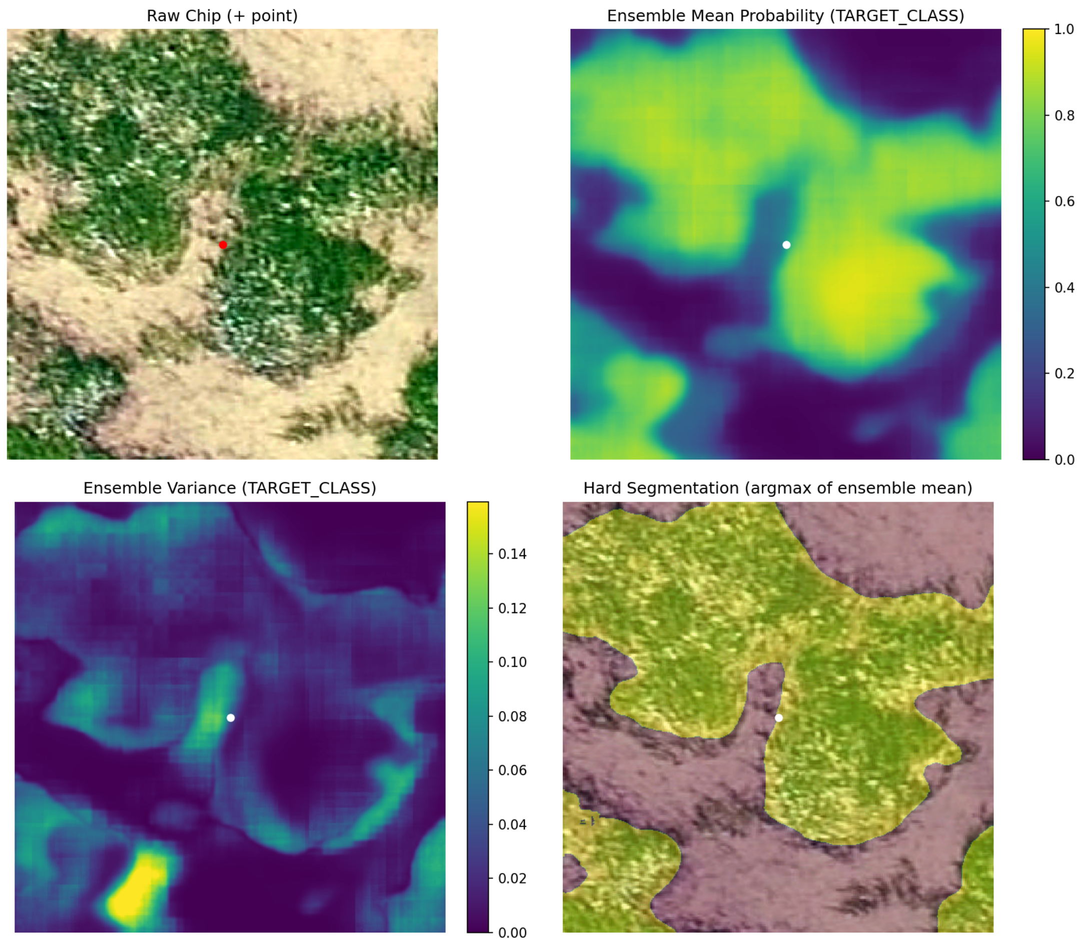


Figure 4.1: Example ensemble inference outputs on a 2022 chip: Raw chip (+ point), ensemble mean probability, ensemble variance, and argmax segmentation.

Table 4.1 summarizes the pixel-wise performance of the ensemble and individual models on the 1,000 points from 2022 for comparison. The heterogeneous ensemble achieves an accuracy of 0.910 and an F-score of 0.886, outperforming all individual members, whose accuracies range from 0.840–0.890 and F-scores from 0.810–0.860. This aligns with previous literature on deep ensembles, which has reported higher accuracy and better-calibrated uncertainty compared to individual models. Another important note is that the precision (0.905) and recall (0.886) are relatively balanced for the deep ensemble.

Table 4.1: Pixel-wise performance of the ensemble and individual segmentation models on the 2022 evaluation points.

Training Year	Model	Precision	Recall	F Score	Accuracy
N/A	Ensemble	0.905	0.869	0.886	0.91
2021	U-Net	0.820	0.890	0.860	0.88
2021	SAM LoRA	0.820	0.890	0.850	0.87
2021	DeepLab	0.880	0.800	0.840	0.88
2018–2021	U-Net	0.880	0.780	0.830	0.87
2018–2021	SAM LoRA	0.810	0.800	0.810	0.84
2018–2021	DeepLab	0.880	0.890	0.860	0.89

To assess how informative the uncertainty measure is, we created selective-accuracy curves on the 2021 in-distribution test set in Figure 4.2. The graph plots 3 metrics: predictive entropy, mutual information, and ensemble variance. For all three metrics, selective accuracy increases sharply as we remove the most uncertain points and then plateaus at an accuracy of about 0.98 for all metrics. This confirms that each metric is meaningfully ordering the errors. The area under the selective accuracy curve (SAUC) is highest for entropy (0.9682) and then closely followed by variance (0.9619), suggesting entropy is the best for ordering the errors. However, as mentioned previously, since entropy is implicitly correlated with the vanilla conformal score and because variance can directly act as a proxy for epistemic uncertainty, growing under drift, we still suggest variance as the metric of choice for creating difficulty adaptive scores.

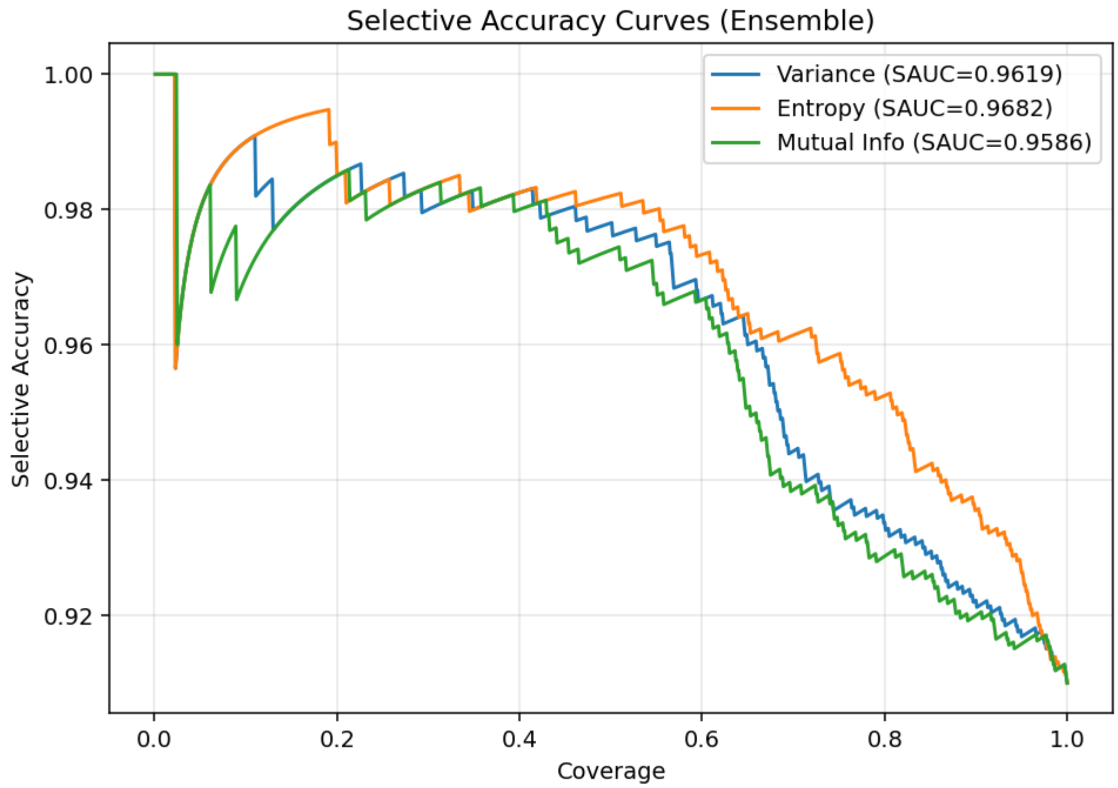


Figure 4.2: Selective accuracy-coverage curves on the 2021 in-distribution test set for entropy-based and variance-based uncertainty scores.

4.2 In-Distribution Evaluation

Table 4.2 displays the coverage and set composition on the 2021 test set for vanilla split CP, linear variance-normalized scores across the λ grid, and the nonparametric normalizer, all at a target $\alpha = 0.1$. As expected, all methods achieved the marginal coverage of 0.9 in 2021. The coverage was tight, as all methods achieved exactly 0.900 coverage. This confirms that the scores were correctly calibrated in distribution.

Table 4.2: In-distribution (2021 test) coverage and set composition for vanilla CP, linear variance-normalized scores across (λ), and the nonparametric normalizer at $\alpha = 0.1$.

Method	q-hat	Coverage	% singletons	% empty	% two-label
Vanilla	0.317	0.9	93.9	6.1	0.0
Linear (λ = 0.5)	0.298	0.9	93.9	6.1	0.0
Linear (λ = 1.0)	0.281	0.9	93.9	6.1	0.0
Linear (λ = 2.0)	0.254	0.9	93.9	6.1	0.0
Linear (λ = 3.0)	0.233	0.9	93.8	6.0	0.1
Linear (λ = 4.0)	0.216	0.9	93.7	6.1	0.2
Nonparametric	1.512	0.9	94.5	4.5	1.0

For the in-distribution evaluation for the linear scale, varying the λ has a minor effect on the set composition. The fraction of singletons remains almost unchanged (from 93.9 to 93.7) from $\lambda = 0$ (vanilla) to $\lambda = 4$ and with only a slight increase in two-label sets (0.2). This stability can be explained by the high concentration of low-variance pixels in the in-distribution 2021 data. As shown in Table 4.3, 92.6% of the pixels of 2021 fall in the lowest variance bin $[0, 0.1)$ and fewer than 1% above 0.3. Since almost all points have a small variance, the linear shrink factor $(1 + \lambda V)$ is relatively close to 1 for the entire test set. Thus, dividing by the $(1 + \lambda V)$ only mildly rescales the original scores without large changes to the ranking of the scores. Since the split CP depends on the score ordering and the threshold is recalibrated for each λ, the same label scores generally lie above or below the conformal threshold, resulting in almost identical set composition.

Table 4.3: Distribution of normalized ensemble variance across bins for 2021 (ID) and 2022 (OOD) evaluation points.

	2021		2022	
Variance bin	N	%	N	%
[0.0, 0.1)	925818	92.6	700	70.0
[0.1, 0.2)	43489	4.3	62	6.2
[0.2, 0.3)	16067	1.6	58	5.8
[0.3, 0.4)	7798	0.8	43	4.3
[0.4, 0.5)	3998	0.4	43	4.3
[0.5, 0.6)	2143	0.2	35	3.5
[0.6, 0.7)	531	0.1	25	2.5
[0.7, 0.8)	110	0.0	9	0.9
[0.8, 0.9)	35	0.0	13	1.3
[0.9, 1.0)	11	0.0	12	1.2

The nonparametric normalizer does produce a slightly different set composition even in-distribution compared to the linear/vanilla scores. The nonparametric normalizer defines its correction parameter using (approximately) equal-size bins on the calibration set, as seen in Figure 4.3. So even within the low variance bin [0,0.1) there are different scale factors. In Figure 4.3, we can see that there is a sharp incline of scale factor within this bin. Thus, the nonparametric method has a more heterogeneous rescaling in the small variance regions, which can slightly alter the score ranking, resulting in a slight change in set composition. While coverage remained at 0.9 in 2021, a slightly different subset of label scores crossed the nonconformity threshold, which thus changes the set prediction of the points.

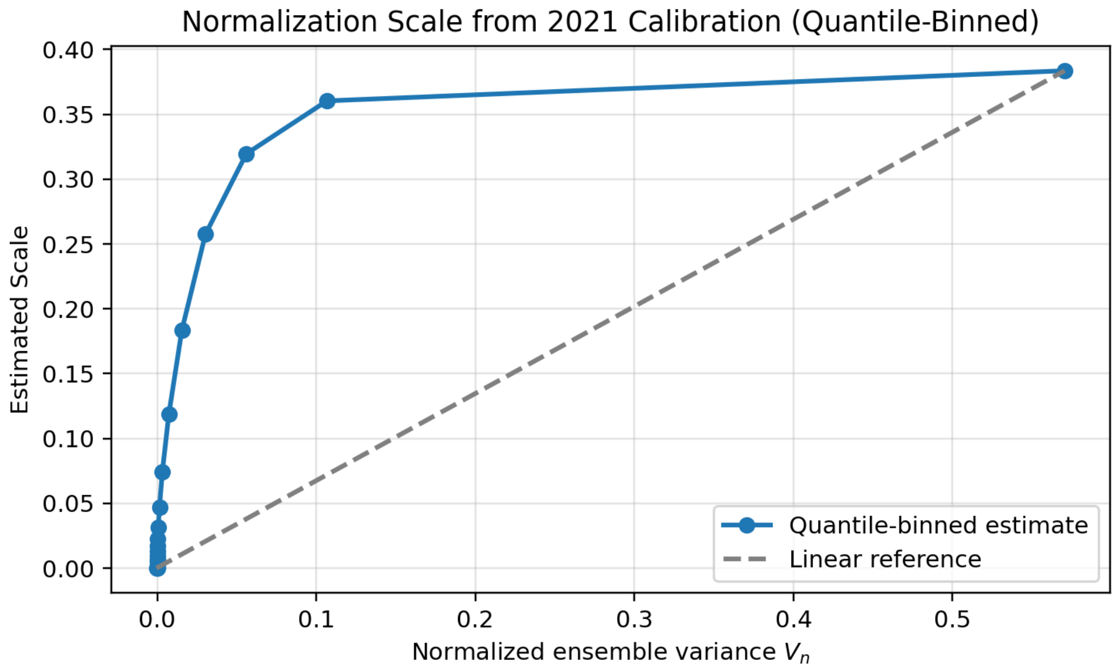


Figure 4.3: Empirical mean nonconformity score (S) versus ensemble variance (V) on the 2021 calibration set, showing the nonparametric scale function.

The main takeaway from the in-distribution results is that variance-aware normalization does not alter marginal coverage for exchangeable data, and as expected, the coverage holds for all methods, as guaranteed by conformal theory. Methods perform almost identically, and differences only become apparent once the exchangeability assumption is violated, which is examined in the following section.

4.3 Temporal OOD (2022)

4.3.1 Linear Normalization

Table 4.4: Temporal OOD (2022) global coverage and set composition for vanilla CP and linear variance-normalized scores across λ at $\alpha = 0.1$.

Method	Coverage	% singletons	% empty	% two-label
Vanilla	0.860	92.7	7.3	0.0
Linear (λ = 0.5)	0.867	94.4	5.6	0.0
Linear (λ = 1.0)	0.871	95.4	4.5	0.1
Linear (λ = 2.0)	0.891	95.6	3.0	1.4
Linear (λ = 3.0)	0.901	93.8	2.5	3.7
Linear (λ = 4.0)	0.908	92.4	2.1	5.5

Table 4.4 and Figure 4.4 show how OOD performance on the 2022 points changes as the linear shrink parameter λ is increased. Vanilla CP undercovers with an empirical coverage of 0.860 for a target of 0.90, reflecting the impact of the temporal drift on the score distribution. As we increase λ, coverage increases monotonically, all the way up to 0.908 at $\lambda = 4$. This comes with the expected trade-off in set composition. The percentage of singletons initially rises, as the percentage of empty sets decreases sharply without a large increase in two-class sets. Singleton percentage then decreases for $\lambda > 2$ as the percentage of two-class sets continues to increase. λ of 3 was chosen as the best choice, as it was the first value to retain target coverage (0.901) while still increasing singletons (93.8%) and with only a modest increase in two class sets (3.7).

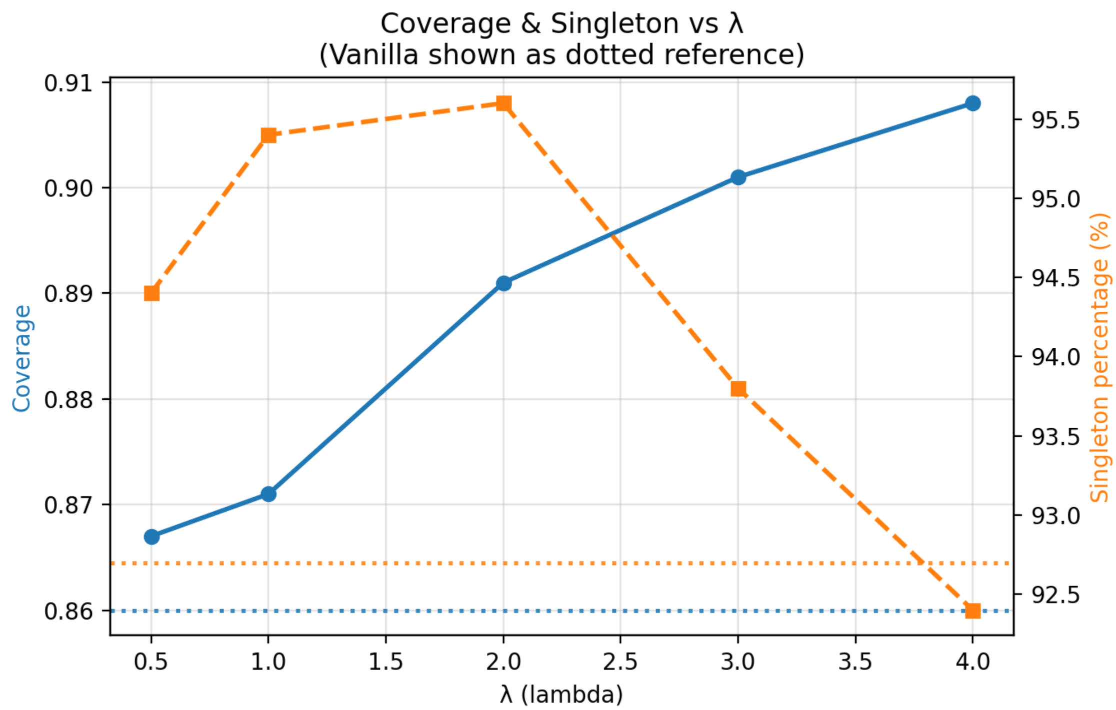


Figure 4.4: Overall coverage and singleton % on 2022 OOD points as a function of the linear shrink parameter (λ) at $(\alpha = 0.1)$.

4.3.2 Nonparametric Normalization

To better match the relationship between nonconformity scores and ensemble variance, as well as remove the need for parameter tuning, we estimated the nonparametric scale function $\hat{a}(V) \approx E[S|V]$ on the 2021 calibration set. Figure 4.5 shows the mean score as a function of variance as shown in Figure 4.3, but with the addition of the empirical results from 2022 for comparison. Average scores increase nonlinearly, increasing rapidly in low variance regions before leveling out. Scores were binned into equal-sized bins, and as seen in Table 4.3, a majority of points in the calibration set were low variance.

The estimated function is able to capture the shape of scores for 2022 shifted data quite well for low variance scores, and although the 2022 curve does deviate from the calibration-derived estimate for higher variance levels, the overall trend remains quite similar. The linear reference shows that the learned normalization is strongly non-linear and cannot be well-approximated with a global linear scaling.

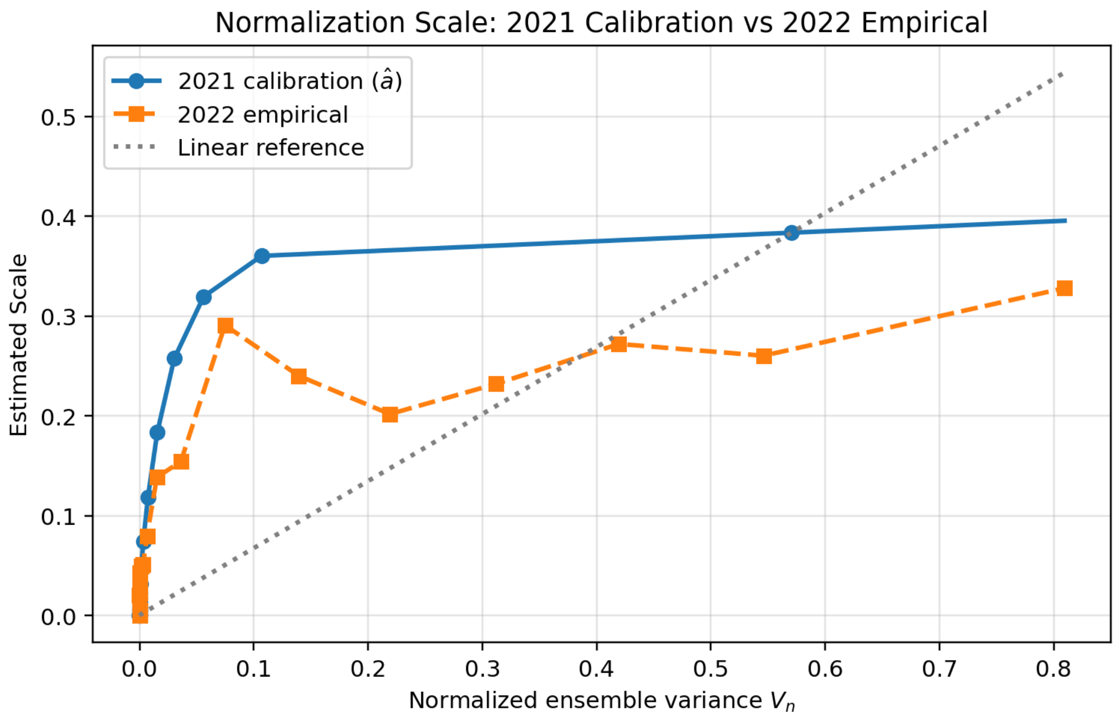


Figure 4.5: Estimated normalization scale from 2021 calibration compared to the empirical score-variance relationship observed in 2022. The extension shows extrapolation beyond calibration support.

Table 4.5 provides a comparison of vanilla CP, the chosen "best" linear method at $\lambda = 3$, and the nonparametric normalizer on 2022 points. While both adaptive methods retain target coverage on the test set, the nonparametric method resulted in a slightly higher coverage of 0.906, making it slightly more conservative at $\alpha = 0.1$. The nonparametric method can do this while also yielding the most favorable set composition with the highest percentage of singletons at 96.4%.

Table 4.5: Temporal OOD (2022) global coverage and set composition for Vanilla CP, Linear normalization ($\lambda = 3$), and the Nonparametric normalizer at $\alpha = 0.1$.

Method	q-hat	Coverage	% singletons	% empty	% two-label
Vanilla	0.317	0.860	92.7	7.3	0.0
Linear (λ = 3.0)	0.233	0.901	93.8	2.5	3.7
Nonparametric	1.511	0.906	96.4	0.8	2.8

The coverage vs variance plot in Figure 4.6 visually compares the three methods and the effect of normalization. The average coverage is calculated for each of the 10 equally sized variance bins (as shown in the figure, the test set resulted in a much wider spread of variance for higher variance regions). Vanilla CP has a fairly steady decline as variance increases, indicating that points with high ensemble disagreement tend to be undercovered. The linear method reduces the decline and even reverses the trend for the high-variance regions, while the nonparametric method results in the most even and closest to target coverage among all variance bins. This suggests the variance-aware normalization is successfully becoming more conservative in difficult or shifted regions.

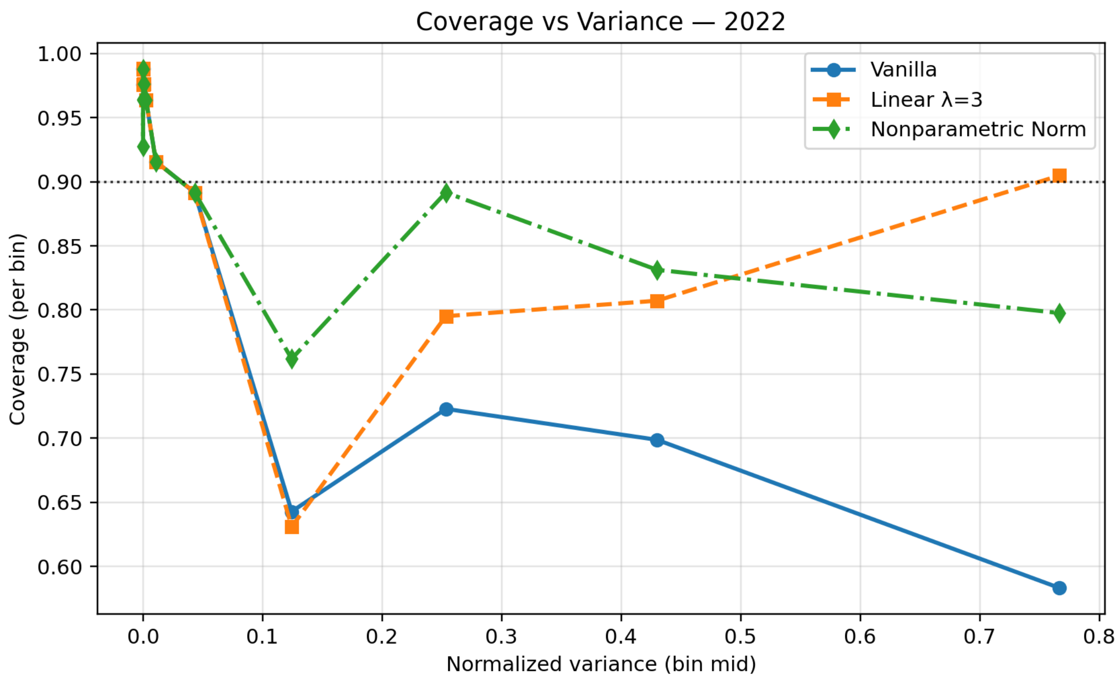


Figure 4.6: Coverage on 2022 OOD points as a function of ensemble-variance bins for vanilla, linear $(\lambda = 3)$, and nonparametric normalization.

4.4 Class Conditional Coverage

Table 4.6 reports the per-class coverage for the 2022 OOD points for the vanilla, linear ($\lambda = 3$), and nonparametric methods. All the methods achieved higher coverage on the background pixels than on eelgrass, reflecting a slight class imbalance. The linear normalization method improves coverage for both classes from vanilla and the nonparametric normalization sees further improvement. Additionally, the linear normalization method improves the disparity in coverage between background and eelgrass classes as it shrinks the percentage difference from 13.4% to 10.4%. The nonparametric method improves further, shrinking the gap to 9.9%. Variance-aware normalization reduces the disparity in the coverage between the two classes.

Table 4.6: Per-class coverage on 2022 OOD points at $\alpha = 0.1$ for vanilla CP, linear normalization ($\lambda = 3$), and the nonparametric normalizer.

Method	Coverage (background)	Coverage (eelgrass)	Difference
Vanilla	0.914	0.780	0.134
Linear (λ = 3.0)	0.943	0.839	0.104
Nonparametric	0.946	0.847	0.099

4.5 Spatial Robustness within 2022

As previously mentioned, spatial robustness is evaluated by grouping the 2022 points into ten spatial blocks based on morton ordering of the coordinates and then computing coverage within each block. Table 4.7 reports the standard deviation of the block-wise coverage and the correlation between the block-wise mean ensemble variance and coverage. As expected, vanilla CP displays the highest block-wise standard deviation in coverage (SD=0.0494) as well as a fairly strong negative correlation between mean variance and coverage (r= -0.622, p=0.0546). Linear normalization reduces spatial variability and weakens the correlation compared to vanilla, while the nonparametric method yields the least variable blocks (SD=0.0357) and the most weakly correlated coverage (r=-0.206, p>0.5).

Table 4.7: Spatial robustness diagnostics on 2022 OOD points: standard deviation of block-wise coverage and correlation between block-wise mean variance and coverage for each method (10 spatial blocks).

Method	SD	Correlation r	p-value
Vanilla	0.049	-0.622	0.055
Linear (λ = 3.0)	0.040	-0.416	0.231
Nonparametric	0.036	-0.206	0.569

Figure 4.7, Figure 4.8, and Figure 4.9 further visualize the summary statistics. Figure 4.7

shows the per-block mean variance to provide an understanding of how much the variance deviates per spatial block. Some areas seem to have consistently higher uncertainty compared to others, such as blocks 2 and 3. Figure 4.8 visualized the per-block coverage for each method. Variance-normalized methods both greatly outperform the vanilla method, and the nonparametric method has the most equal performance, with the majority of blocks being close to the target coverage. Figure 4.9 visualizes the correlation between the block-wise coverage and mean variance, showing how the nonparametric method specifically reduces the negative correlation between variance and coverage.

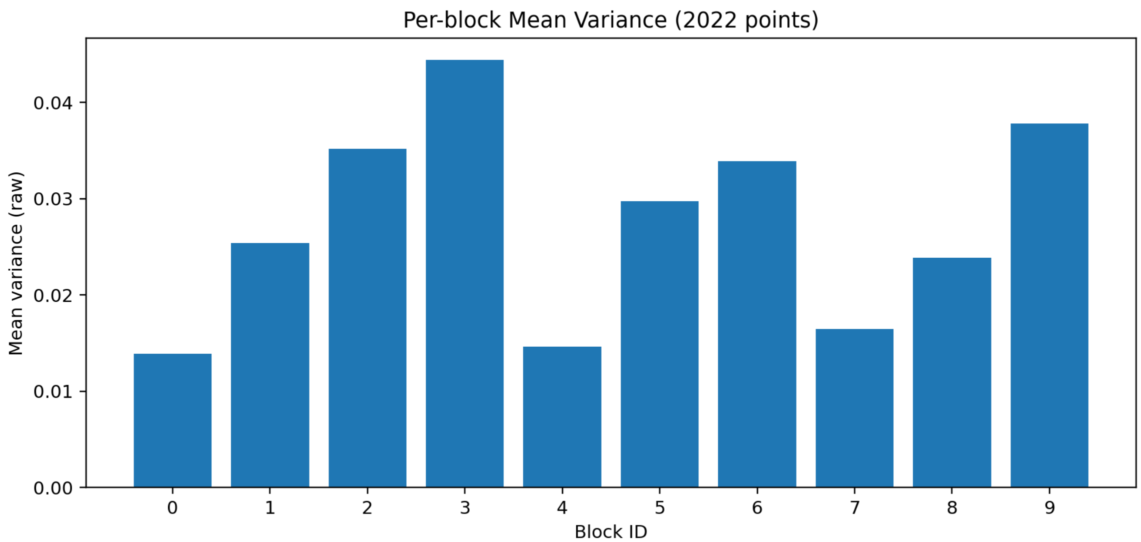


Figure 4.7: Per-block mean variance for 2022 points

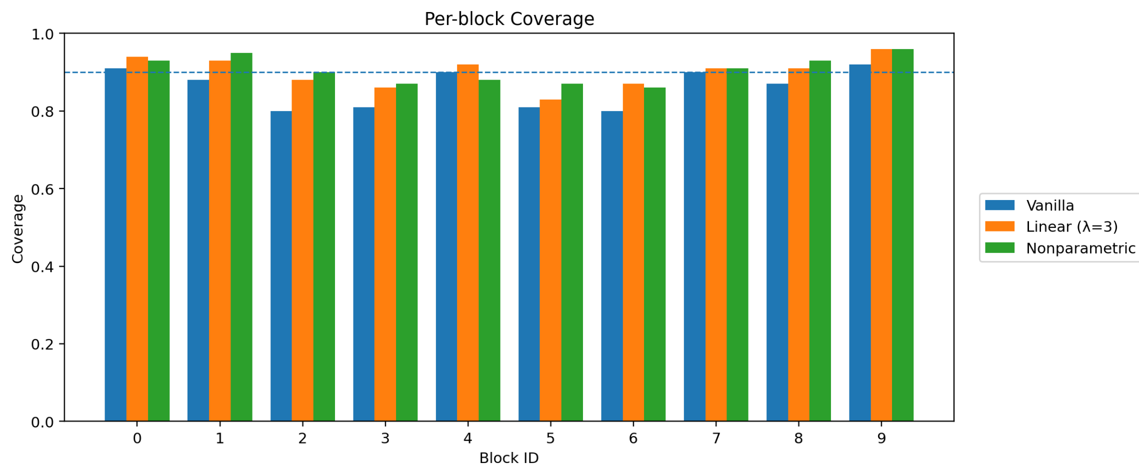


Figure 4.8: Per-block coverage for 2022 points for vanilla, linear $(\lambda = 3)$, and nonparametric methods

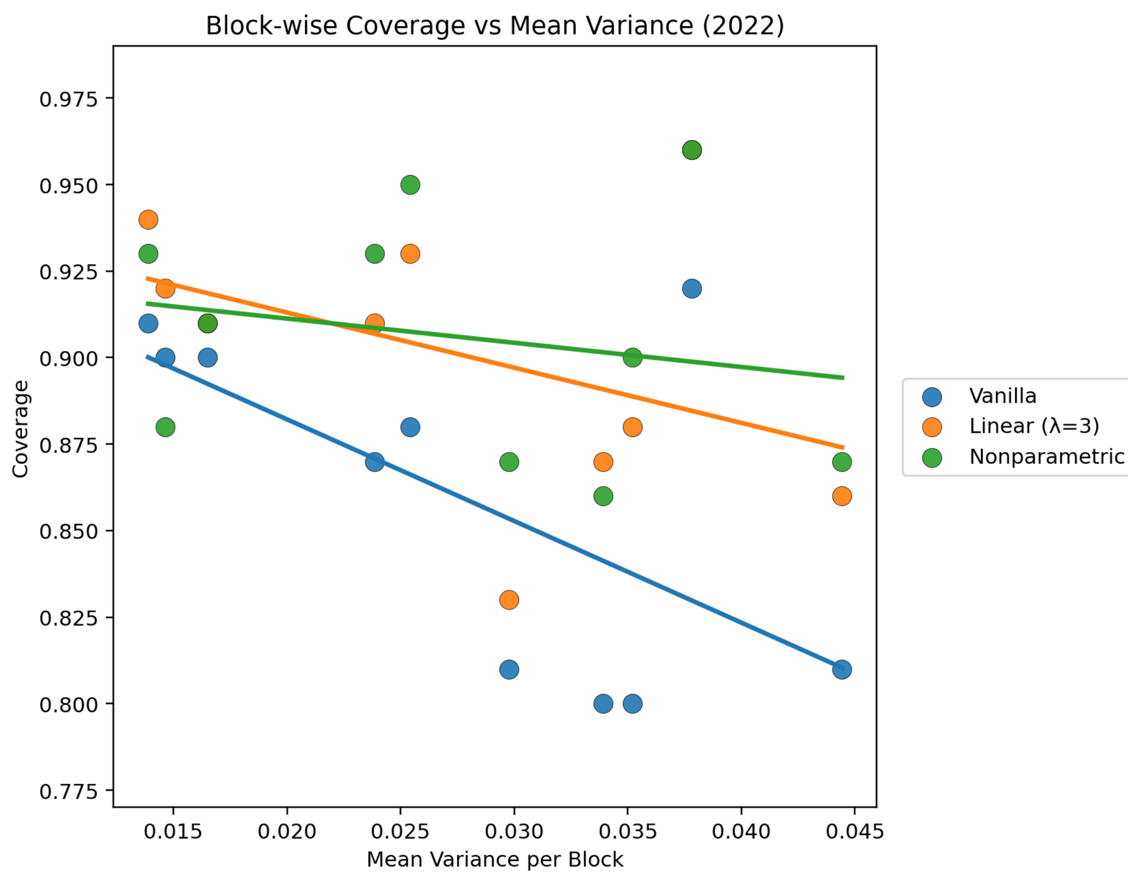


Figure 4.9: Scatter plot of block-wise coverage vs mean variance for vanilla, linear $(\lambda = 3)$, and nonparametric methods

4.6 Visual Method Comparison

Finally, Figure 4.10, Figure 4.11, and Figure 4.12 show the actual raster and overlaid predictions. Figure 4.10 shows the 2022 raster as well as the ground truth points. As seen in the figure, the majority of eelgrass points are centered in the middle of the estuary imagery.

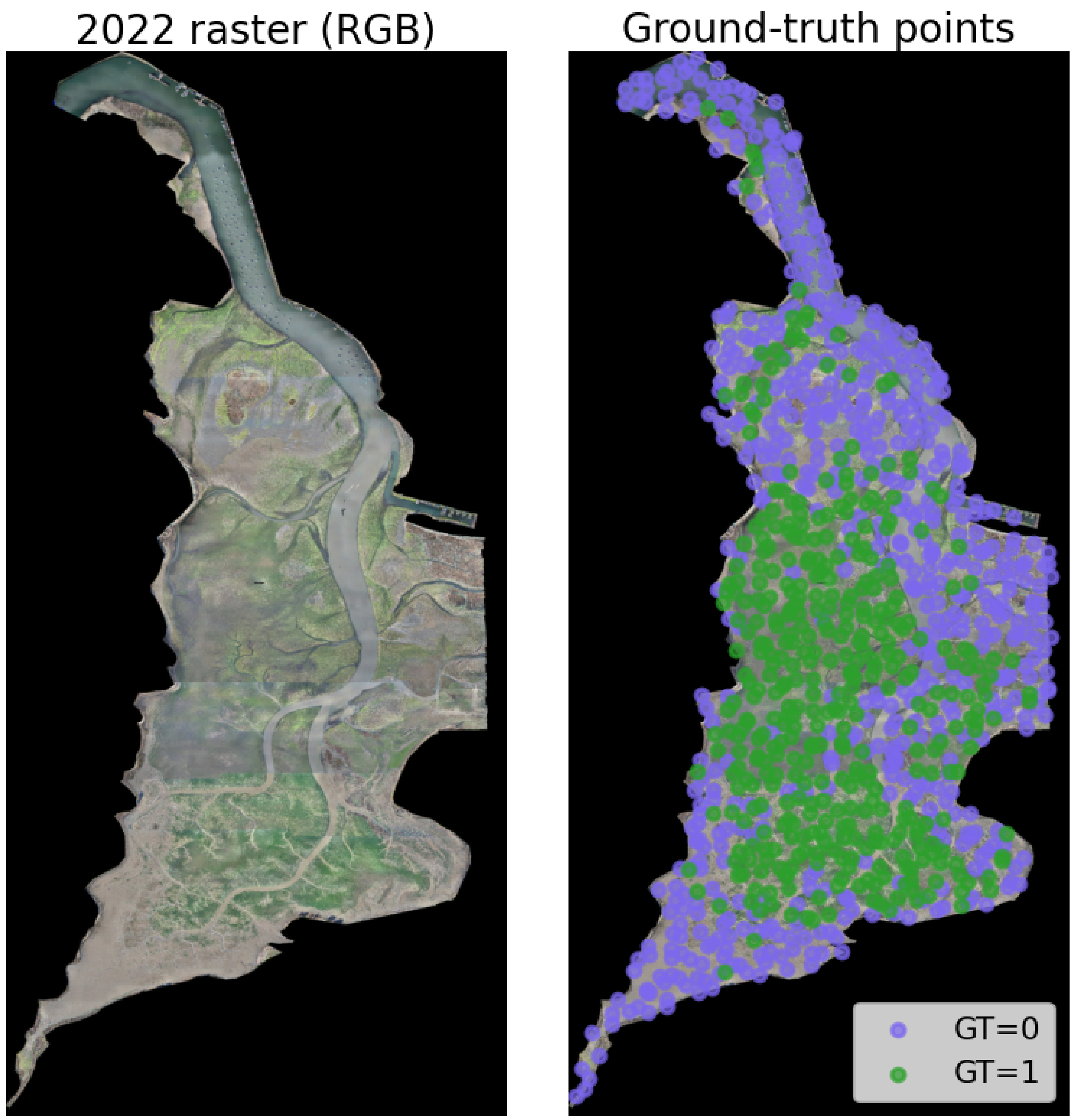


Figure 4.10: Visuals of the 2022 raster (left) and overlayed ground truth points (right) - Blue = Other, Green = Eelgrass.

Figure 4.11 displays the set composition for vanilla, linear, and nonparametric methods. As seen in the image, there appears to be a high uncertainty region towards the top end of the estuary. For this region, the vanilla CP method reported a majority of incorrect singletons or empty sets, while the normalized methods converted many of these into two-class sets. The vanilla method has quite a few empty sets and almost no two-set classes. The linear method reduces the number of empty sets, and the nonparametric method reduces this number further, with some being converted to singletons and some two-set classes.

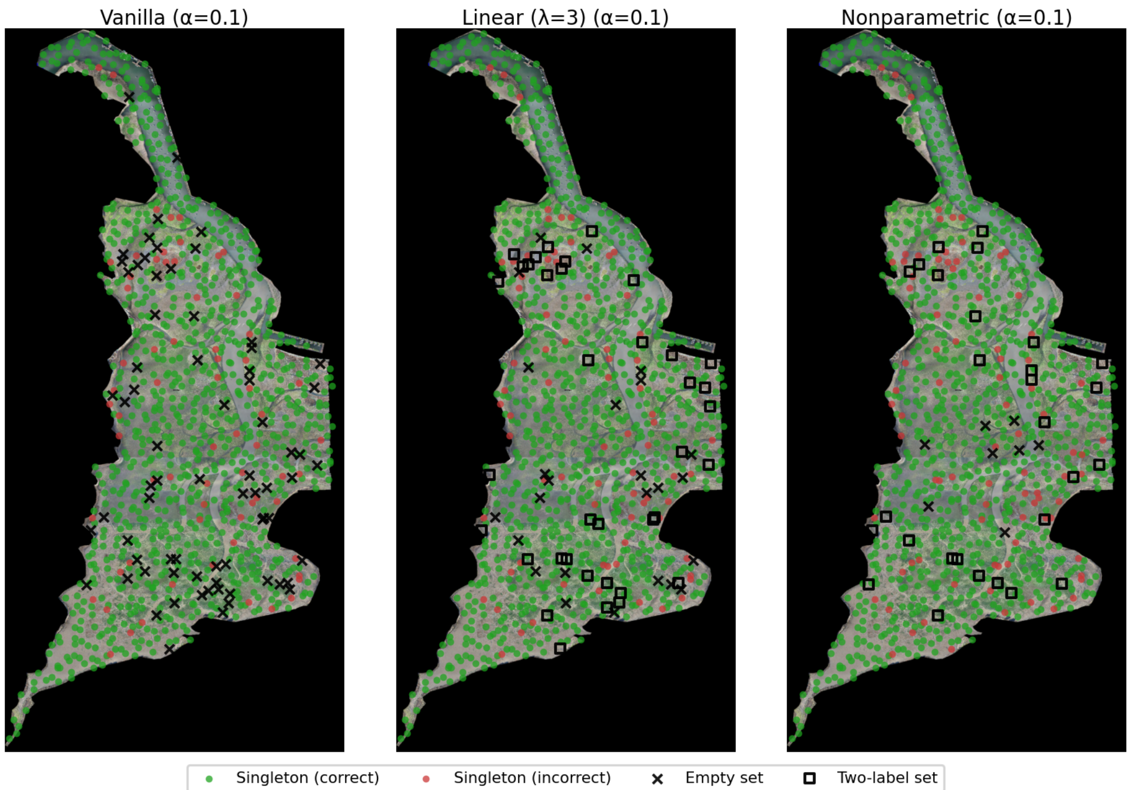


Figure 4.11: Set composition overlay for 2022 raster for vanilla, linear ($\lambda = 3$), and nonparametric methods.

Figure 4.12 provides a simplified view of Figure 4.11 by simply visualizing how the conformal prediction set composition changes when moving from the vanilla method to each variance-normalized alternative. Points are categorized by the type of transition; teal squares indicate where predictions expand to two-label sets (from either empty sets or incorrect singletons), green circles indicate empty sets that became correct singletons, red circles indicate empty sets to incorrect singletons, and black points indicate any other type of transition (uncommon).

For both methods, many of the two-set class transitions occur in the same relative area, indicating both methods consistently identify similar high-difficulty areas of the estuary. Compared to the linear normalizer, the nonparametric method produces a larger number of empty-to-singleton transitions. Both methods have a similar increase in two-label predictions in areas where the vanilla method was overconfident and wrong.

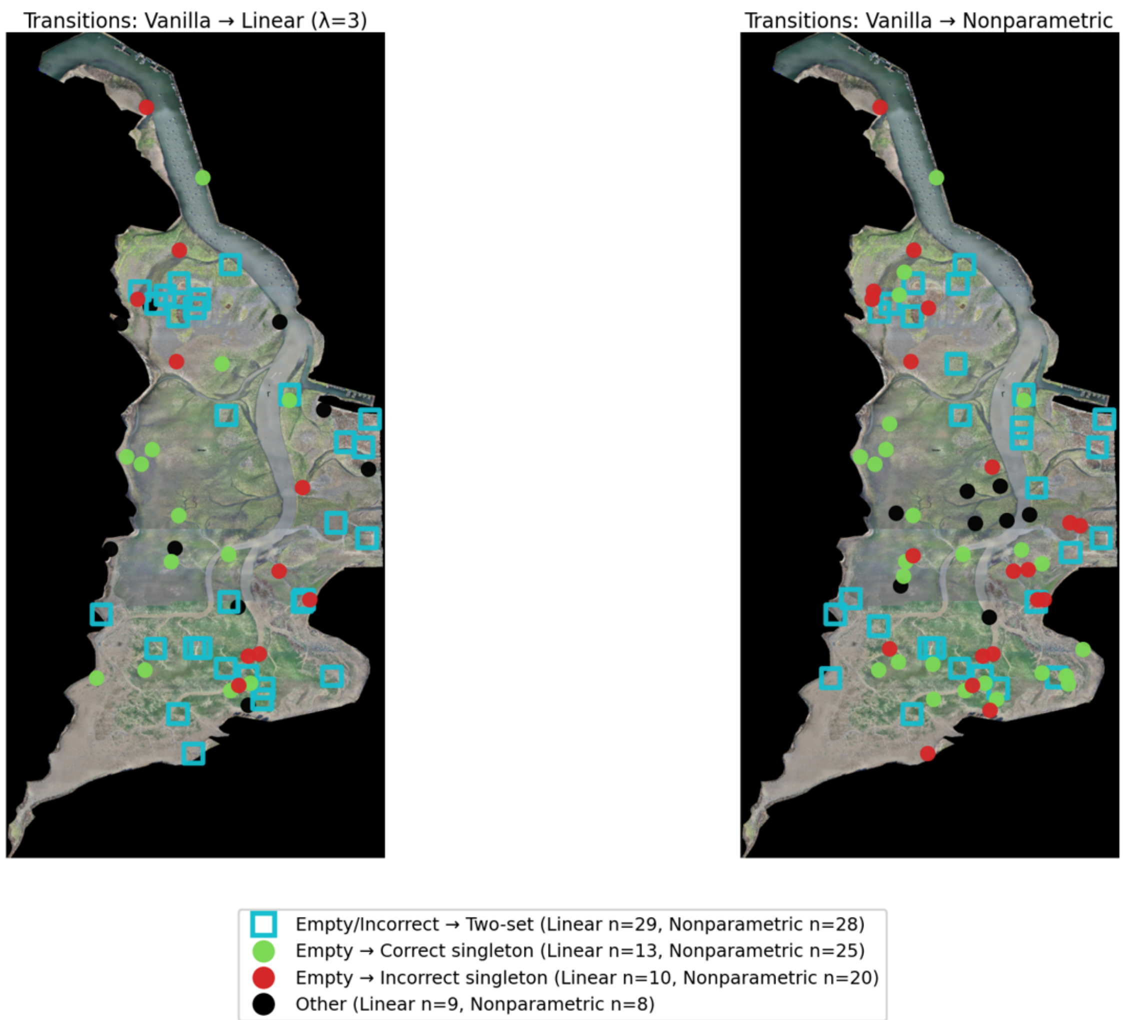


Figure 4.12: Set composition changes from vanilla for linear $(\lambda = 3)$ and nonparametric methods. Counts for each transition type are reported in the legend.

Figure 4.13 provides a localized view of how the three methods differ, zooming in on a single 448x448 px chip (same chip from Figure 4.1). It provides a good example of how variance normalization can change the predictions. At the labeled point, the ensemble mean predicts eelgrass with low confidence ($\hat{p} \approx 0.56$), yet the corresponding ensemble variance is also fairly low ($\hat{v} \approx 0.045$). This is a case where predictive uncertainty is relatively high (pixel is intrinsically ambigious, so we have high entropy/low confidence) but epistemic uncertainty is low (the ensemble members agree, resulting in low variance). As a result, both vanilla CP and linear variance-normalized method produce an empty set prediction at the labeled point. In contrast, the nonparametric method, retains eelgrass as the correct singleton set prediction. This shows an example of how the nonparametric function can better account for the typical score distribution at a certain variance level (in Figure 4.3, expected score quickly rose for small variance and then stabilized). We can see these results even more clearly in the entire segmented chip, vanilla CP results in a large portion of empty sets in areas of ambiguity (borders between eelgrass and other as well as darker shadows). The linear method is able to shrink these ambiguous areas to singletons and two-set classes (as seen by the thinner strips between other and eelgrass) but still has a large portion of empty set predictions. Nonparametric is able to further shrink the area of ambiguity, while also producing almost no empty sets, instead providing more informative 2 class sets.

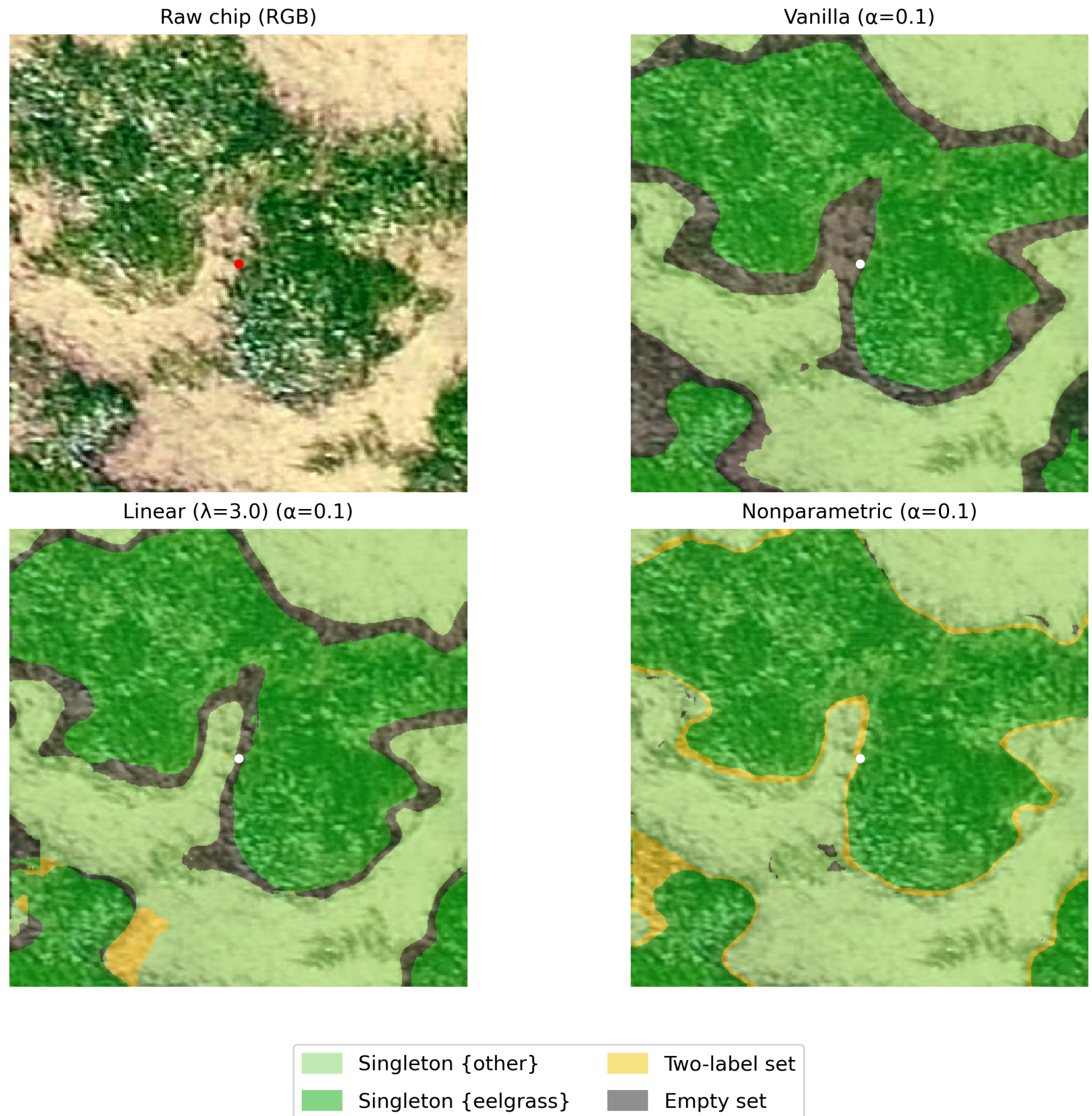


Figure 4.13: Chip-level set prediction comparison on a representative 2022 point (GT = eelgrass). Both vanilla and linear methods return empty sets at that pixel, while the nonparametric method returns the correct singleton {eelgrass}.

4.7 Sensitivity Analysis

Lastly, we examined how the methods performed across different target miscoverage levels ($\alpha \in \{0.10, 0.05, 0.025\}$). Table 4.8 reports the coverage and set composition for each method and α. Figure 4.14 provides a visual of the overall coverage trend for each method, while Figure 4.15 provides a comparative visual of the set compositions.

Table 4.8: Coverage and set composition on 2022 OOD points across $\alpha \in 0.10, 0.05, 0.025$ and methods.

Method	Coverage	Empty (%)	Single (%)	Two-label (%)
α = 0.025				
Linear (λ=3)	0.967	0.0	70.8	29.2
Nonparametric	0.972	0.3	65.4	34.3
Vanilla	0.949	0.0	87.2	12.8
α = 0.05				
Linear (λ=3)	0.952	0.0	80.5	19.5
Nonparametric	0.944	0.4	87.0	12.6
Vanilla	0.919	0.0	95.9	4.1
α = 0.1				
Linear (λ=3)	0.901	2.5	93.8	3.7
Nonparametric	0.906	0.8	96.4	2.8
Vanilla	0.860	7.3	92.7	0.0

Across all α levels, the main pattern seen at $\alpha = 0.01$ continues to be true. Vanilla CP largely undercovers, while the variance-aware normalization methods can move the coverage closer to or slightly above the target coverage. The nonparametric method appears to perform best for both $\alpha = 0.1$ and $\alpha = 0.025$, while the linear method is the only method to exceed the target coverage for $\alpha = 0.025$.

As expected, the variance-aware normalization methods both increase the number of two-set classes compared to vanilla CP, a necessary tradeoff to achieve target coverage. However, the methods are not providing overly conservative results, as the empirical coverages were close to the targets, with no extreme inflation of two-class sets. The methods produce conservative estimates when needed, flagging areas where two-set class predictions are necessary.

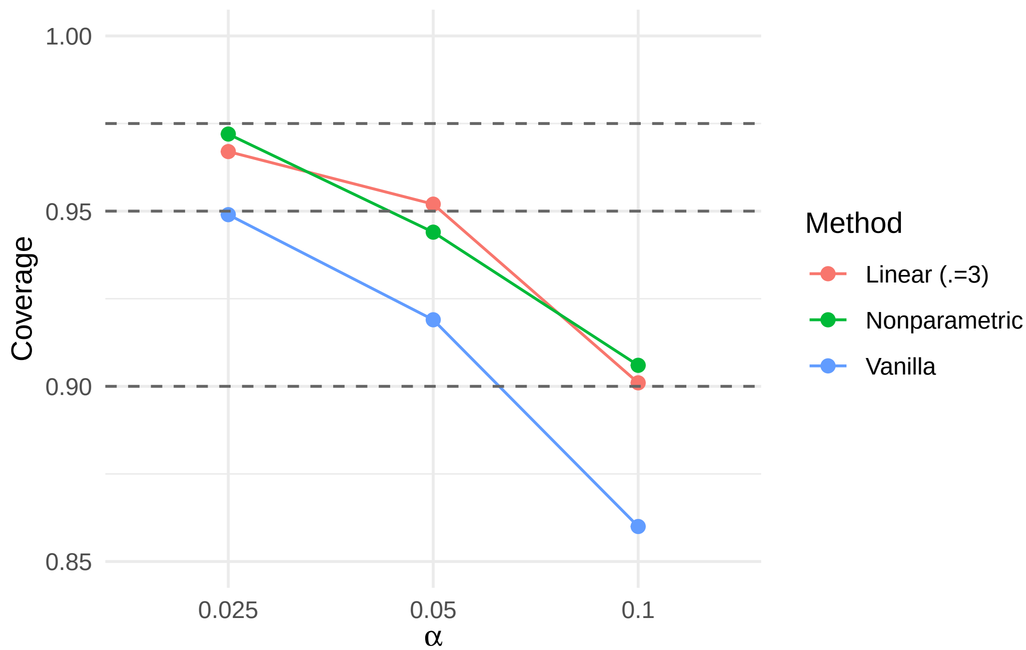


Figure 4.14: Overall coverage on 2022 OOD points across $\alpha \in \{0.1, 0.05, 0.025\}$ for each method. Dashed line shows the target coverage 1 - α.

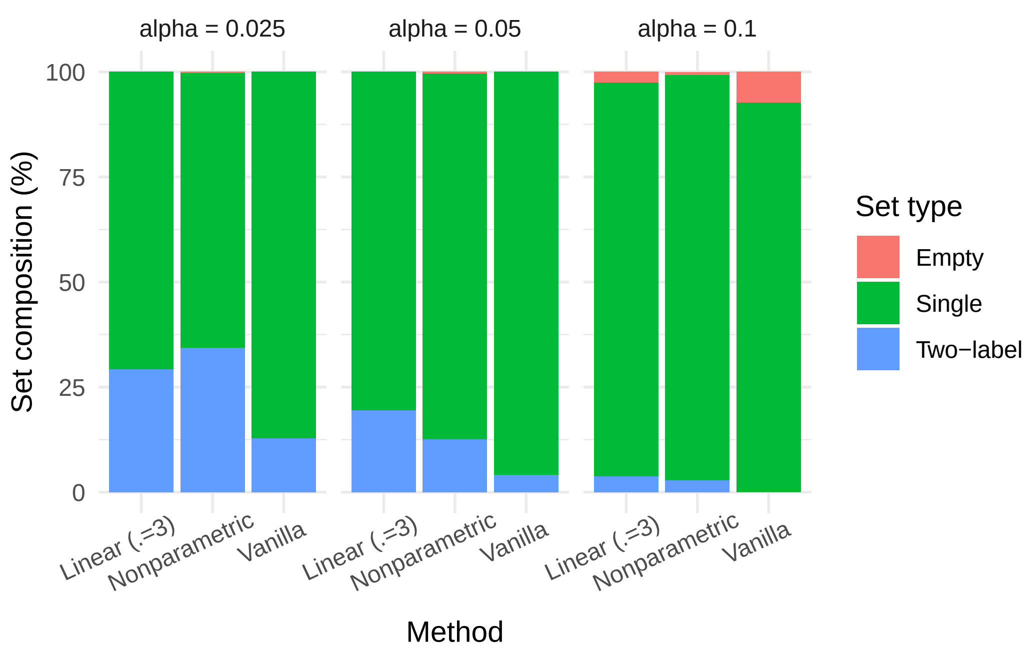


Figure 4.15: Set composition (empty, single, two-label) for each method on 2022 OOD points across $\alpha \in \{0.1, 0.05, 0.025\}$.

Chapter 5

CONCLUSION

This study developed and evaluated a practical uncertainty pipeline for time-series eelgrass segmentation that is reliable under temporal distribution shift. We were motivated by the reality that ecological modeling is deployed across years with differing image conditions and requires useful uncertainty estimates for truly reliable models. Building off existing strong deep segmentation models, we combined heterogeneous deep ensembles, probability calibration, and conformal prediction to produce classification sets at a user-defined confidence level. While standard conformal prediction provides finite-sample marginal coverage under exchangeability, realistic year-over-year imagery often violates this assumption, causing conformal prediction to under-cover on temporally shifted data and resulting in a larger concentration of errors in the most difficult regions, precisely where uncertainty estimates would be most useful.

The primary contribution is an investigation of variance-aware normalization strategy for conformal classification under temporal distribution shift. We normalized the vanilla nonconformity score by ensemble disagreement (member-probability variance after temperature scaling) as a proxy for epistemic difficulty. We specifically studied two variants: a parametric linear normalization $S/(1 + \lambda V)$ and a nonparametric normalization $S/\hat{a}(V)$ with $\hat{a} \approx E[S|V = v]$ learned via quantile binning and interpolation on the calibration set. The approach is based on score normalization ideas in conformal prediction, which have not been thoroughly investigated under drift.

On Morro Bay imagery, the ensemble improved predictive performance compared to the individual models, and the produced uncertainty scores effectively ordered errors. In distribution (2021), all conformal methods achieved target coverage on a held-out test set with similar set compositions, which confirms that the normalization does not degrade coverage under exchangeability. Under temporal shift (2022), vanilla conformal prediction undercovered, with errors, specifically concentrated on areas of high uncertainty. Variance-aware normalization was able to largely mitigate the issue and recover the lost coverage. The linear method recovered nominal coverage under a moderate λ of 3, resulting in a reduction of empty sets and a slight increase in 2-set classes. The nonparametric method performed the best overall, achieving near or slightly conservative coverage for all α and best set composition. Beyond the global coverage, variance-aware normalization also improved the class-conditional coverage disparity by reducing the gap between eelgrass coverage and other. It also improved the spatial robustness by lowering blockwise coverage variability and weakening the relationship between blockwise coverage and uncertainty.

Overall, the results support the claim that a shift-aware epistemic proxy can be used to stabilize the conformal scores under a slight temporal shift without retraining, test-time labels, or any additional models training beyond those trained during model selection. This framework can be especially useful for environmental monitoring as it provides users with both predictions and an estimate of reliability. It flags areas of uncertainty, where future annotations would be most impactful.

5.1 Limitations

There are several limitations to be considered with the conclusion. First, conformal coverage guarantees are formally tied to exchangeability. While variance-aware normalization improved the empirical robustness under drift, it does not provide a theoretical guarantee. The approach relies on the assumption that the relationship between score magnitude and the difficulty proxy is stable across years, which could fail under larger shifts. Additionally, the study evaluates temporal OOD behavior using point-labeled 2022 data, as full polygon data labeling can be quite extensive. This can be useful, but could miss spatial patterns in fully segmented tiles. Lastly, the uncertainty metric is approximated via ensembles. Although this is a common method for approximating uncertainty, it also introduces additional computational cost.

5.2 Future Work

Several possible areas remain for extending and strengthening this work. While variance-aware score normalization demonstrated strong empirical performance under moderate shift, future work should benchmark this approach against other conformal methods designed for distribution drift, such as weighted conformal prediction. Comparisons on high-dimensional imagery could provide useful insight into how conformal approaches scale. Future work could also examine robustness under larger distribution shifts, including imagery from different seasons, or entirely different estuaries. This would help provide understanding on whether the learned relationship between conformal scores and epistemic difficulty proxies can remain stable under stronger domain changes. Future work could also explore alternative uncertainty proxies.

Lastly, theoretical work could investigate the behavior of normalizing scores under non-exchangeable data. Understanding how scores normalization interacts with covariate shift could help clarify when these strategies prove to be most effective.

5.3 Computational Details

All experiments and analyses were implemented in Python and executed in a Windows ArcGIS Pro Python environment. Deep-ensemble inference was run via arcgis.learn (loading Esri .emd models) and PyTorch, using an NVIDIA GPU. Geospatial inputs were handled using Rasterio, Fiona, Shapely, and PyProj; numerical computing and tables used NumPy and pandas; statistics utilities used SciPy and scikit-learn; figures were generated with Matplotlib (with tqdm for progress logging). Code to reproduce the computational workflow is available at https://github.com/DillonMurphy04/eelgrass-cp. Package dependencies are listed in requirements.txt, and full documentation can be found in the repository README.

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